

The topology *is* the walk basis

Generated architectures (Petersen, Mycielski, Steiner) and the walk-geometry of structural control

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Abstract

The StructuralActor (SA-HSiKAN) controller does not message-pass; it gathers along the topology’s **signed walk-holonomy** $B^L = (A_+ - A_-)^L$. So the interconnection graph does not sit *beside* the computation — it **generates** it: the walks are the structural prior. This reframes “which topology controls which plant” (Kato’s isomorphic-controllers program) as *which walk geometry* controls. We add the extremal generated architectures — Petersen, Mycielskian, Kneser, random-regular expander — to the zoo and compute the graph invariants that are, read correctly, **walk-structure measures**. The finding: **Petersen, Steiner $S(2, 3, 9)$, and Fano $S(2, 3, 7)$ all reach frame coherence 0.33 — the tightest in the zoo** — meaning their walks are maximally distinct (an equiangular tight frame), reached by two independent routes: combinatorial *design* and spectral *expansion*. Hub/sunflower topologies sit at coherence 1.0 (degenerate). **But a matched- N control sweep refutes the tempting prediction:** the tight-frame walk geometry is *not* a universal control optimum — Petersen is near-worst on average and coherence does not rank control. The real law is **matching**: every plant is best controlled by its *own* topology (9/9), so structure is load-bearing and an isomorphic controller wins, but there is no one-topology-fits-all.

1 Walks, not message-passing

SA-HSiKAN replaces HSiKAN’s iterated signed message-pass with one precomputed operator $B^L = \text{matrix_power}(A_+ - A_-, L)$ — the signed L -hop adjacency power, i.e. the sum over length- L walks of the product of edge signs (the $\mathbb{Z}_2$ holonomy). The controller’s structural content is therefore *exactly* the set of walks the topology admits. Designing the graph = designing the walk basis. This is why the choice of generated architecture is load-bearing rather than cosmetic.

2 The invariants are walk-structure measures

For each topology we compute four quantities; each is a property of the *walks*:

invariant	walk reading	extremal case
frame coherence	overlap of different start-nodes’ walks	low = distinct walks (tight frame)
adjacency spectral gap	how fast walks mix	high = rapid mixing (expander)
$\mathbb{Z}_2$ frustration	holonomy around walk-cycles	0 = balanced (consistent transport)
girth / degree-regularity	shortest returning walk / uniformity	Petersen girth 5, 3-regular

3 The zoo, measured

Extremal generated architectures added through the existing `topology_zoo.signed_graph` (edge-lists), verified by their defining invariants (Petersen is strongly-regular $(10, 3, 0, 1)$; Kneser $K(5, 2) \cong$ Petersen; Grötzsch $= \mu(C_5)$ is triangle-free). Invariants at the zoo’s natural sizes:

topology	alg. conn.	spec. gap	coherence	frustration	deg. irreg.
petersen	2.00	2.00	0.33	2	0.00
steiner9	1.70	1.73	0.33	—	0.49
fano7	1.59	1.59	0.33	0	0.00
ring	0.47	0.47	0.50	0	0.00
kneser(9, 2)	20.0	20.0	0.52	—	0.00
complete	9.00	9.00	0.62	11	0.00
random / expander	0.6	1.6/0.6	0.67	1/2	0.9/0.0
grotzsch ($\mu(C_5)$)	2.23	2.70	0.67	3	0.64
chain / grid / kuniform3	0.1–1.0	0.3–1.4	0.71	0–2	0.4–0.9
star / tree / sunflower	0.2–1.0	0.5–2.8	1.00	0	0.9–2.2

Table 1: Coherence (walk overlap) is the discriminating axis. The tight-frame cluster (green in the figures) — Petersen, Steiner, Fano — reaches 0.33; the hub/sunflower degenerate cluster reaches 1.0.

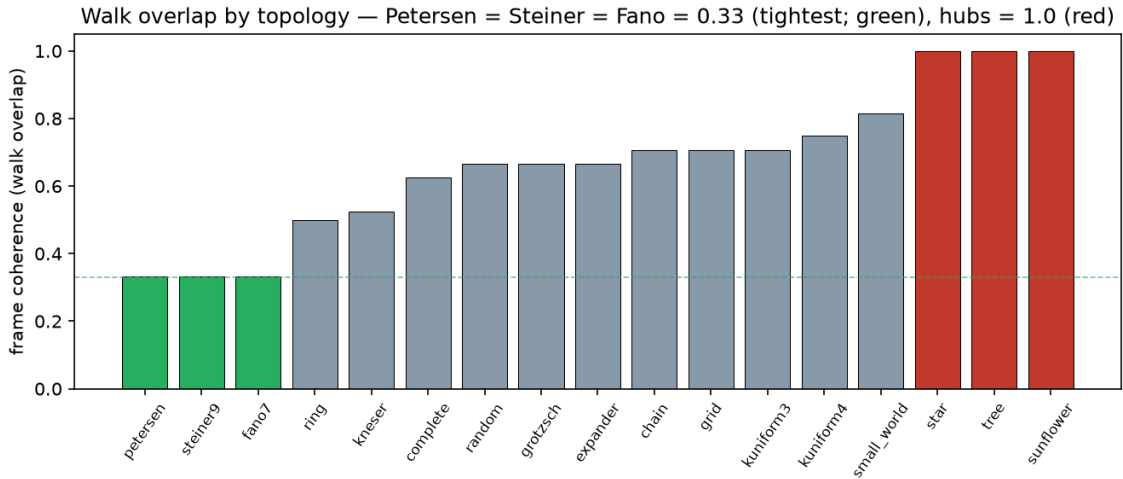


Figure 1: Walk overlap by topology. Petersen = Steiner = Fano = 0.33 (tightest, green); the hubs = 1.0 (red).

4 The finding: matching, not a universal walk geometry

Two routes to the same walk frame (structural). A balanced 2-design (Steiner/Fano — every pair of points in exactly one block) and a strongly-regular expander (Petersen — $\lambda=0, \mu=1$) both produce an equiangular tight frame of walks: coherence $1/3$, *sign-independently* (every pair shares exactly one common neighbour/block). Mycielski adds a third, hierarchical route (coherence

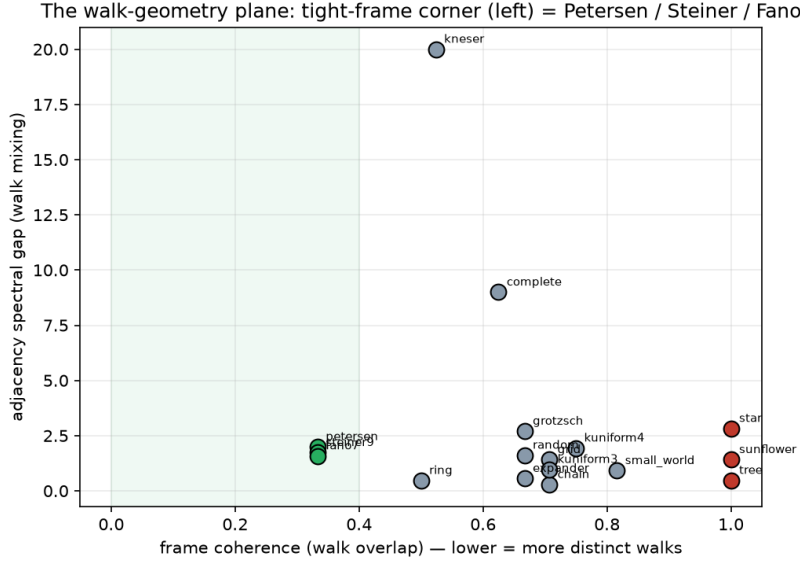


Figure 2: The walk-geometry plane (spectral gap vs. coherence). The tight-frame corner (low coherence) is reached by both the combinatorial designs (Steiner, Fano) and the spectral expander (Petersen).

0.67, frustration 3). The tempting prediction was that this tight-frame geometry is the best control basis.

The matched- $N=10$ control sweep refutes it. Each topology, instantiated as a learned HSiKAN controller, fitting each plant’s structural target (median over seeds):

controller	avg. MSE over plants (lower better)	coherence
chain	0.391 (best)	0.71
expander	0.404	0.67
random	0.416	0.58
tree	0.423	1.00
...	...	...
petersen	0.542 (7th / 9)	0.33
ring / star	0.563 / 0.565	0.50 / 1.00

Frame coherence does not predict control — Petersen, the tightest frame, is near-worst; chain is best; no monotone relation (Fig. 3). The tight-frame walk geometry is *not* a universal control optimum.

What the sweep *did* reveal is stronger — a matching law. The best controller for every one of the nine plants is its *own* topology (best[*plant*] = *plant*, 9/9). So **structure is load-bearing and the matched (isomorphic) controller wins** — the controller’s walk-frame should be isomorphic to the plant’s — but there is no one-topology-fits-all. The earlier “Steiner is the best basis” was therefore *plant-specific*, not a tight-frame law.

5 Status and reading

Generators, invariants, and the topology→control map are implemented and tested (`topology_zoo`, `topology_invariants`, `controller_bench`). **Reading:** the walk-as-structural-prior picture holds

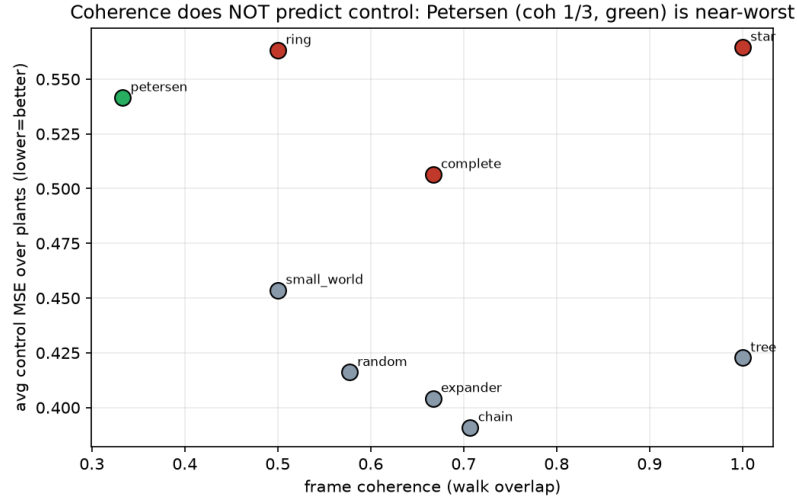


Figure 3: Coherence vs. average control MSE: no relationship. Petersen (coherence 1/3, green) is near-worst; chain (coherence 0.71) is best.

(the matched walk-frame wins), but the universal-tight-frame prediction does *not* (coherence does not rank control). The isomorphic-controllers law is **matching**, not a single best topology — a cleaner, falsifiable result. *Caveat*: this is the supervised structural-fit (Phase 1, 2 seeds); the closed-loop confirmation on real plants is the natural follow-up. Plan: docs/plans/2026-06-28-extremal-topology-zoo